Unit- 3
Library
Functions

Library Functions

Library functions are pre-defined functions provided by Arduino libraries that simplify programming tasks.

1. I/O Functions (Input/Output Functions)
2. Serial Communication Functions

I/O Functions (Input/Output Functions)

These functions are used to interact with sensors and actuators.

Function	Description
pinMode(pin, mode)	Configures a pin as INPUT, OUTPUT, or INPUT_PULLUP
digitalWrite(pin, value)	Sets a digital pin HIGH or LOW
digitalRead(pin)	Reads the state of a digital pin
analogWrite(pin, value)	Outputs PWM signal
analogRead(pin)	Reads analog value from a sensor

```
void setup() {
  pinMode(13,
OUTPUT);
}

void loop() {
  digitalWrite(13, HIGH);
  delay(1000);
  digitalWrite(13, LOW);
  delay(1000);
}
```

Serial Communication Functions

Serial communication allows Arduino to exchange data with a computer or another device.

Function	Description
Serial.begin(9600)	Starts serial communication
Serial.print()	Prints data without a new line
Serial.println()	Prints data with a new line
Serial.read()	Reads incoming serial data
Serial.available()	Returns available bytes in buffer
Serial.write()	Sends binary data

```
void setup() {
  Serial.begin(9600);
}

void loop() {
  Serial.println("Hello IoT");
  delay(1000);
}
```

IoT Communication Protocols

Communication protocols define how IoT devices exchange d

MQTT (Message Queuing Telemetry Transport)

Lightweight protocol
Uses Publisher-Subscriber model
Suitable for low-bandwidth networks
Commonly used in IoT applications

HTTP (HyperText Transfer Protocol)

Used for web communication
Client-Server architecture
Common in web-based IoT applications

CoAP (Constrained Application Protocol)

Designed for resource-constrained devices
Similar to HTTP but lightweight
Uses UDP instead of TCP

Bluetooth

Short-range wireless communication
Low power consumption
Used in wearable devices

ZigBee

Low-power wireless protocol
Mesh networking support
Suitable for home automation

Wi-Fi

High-speed wireless communication
Provides Internet connectivity

Basic IoT Network Topologies

Topology refers to the arrangement of devices in a network.

A. Star Topology
B. Bus Topology
C. Ring Topology
D. Mesh Topology
E. Tree Topology

Basic IoT Network Topologies

A. Star Topology

- All devices connect to a central hub or gateway.

Advantages

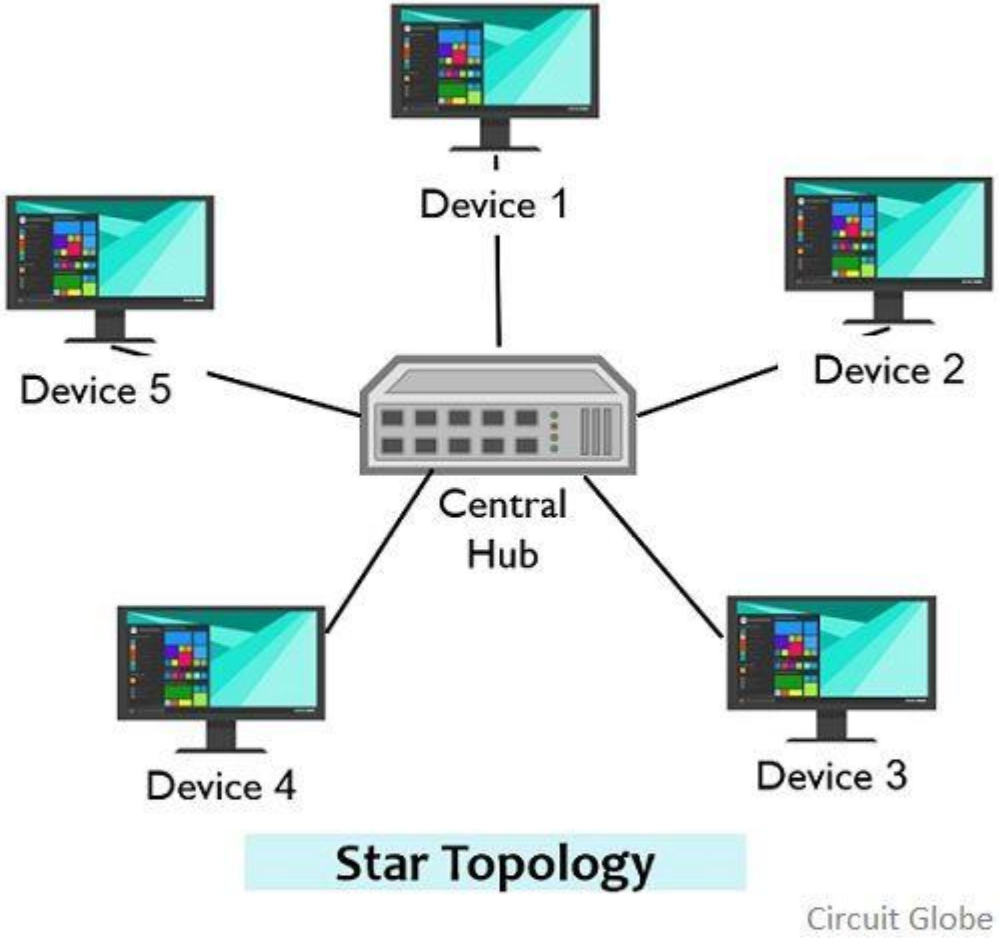
- Easy management
- Easy fault detection

Disadvantages

- Central node failure affects entire network

Example

Wi-Fi router connected to multiple IoT devices.



B. Bus Topology

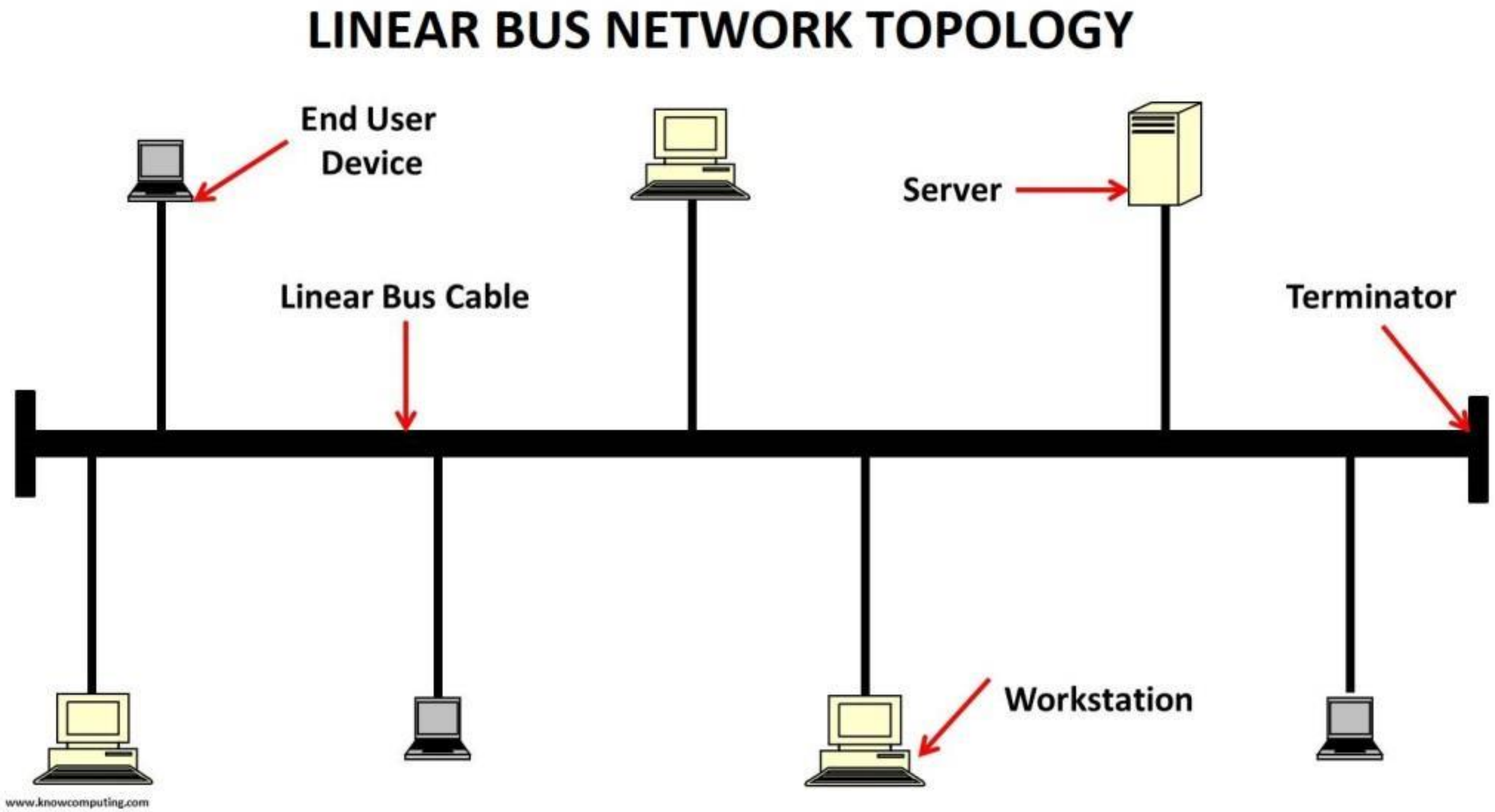
- All devices share a common communication line.

Advantages

- Simple design
- Low cost

Disadvantages

- Failure of main cable affects communication



Basic IoT Network Topologies

C. Ring Topology

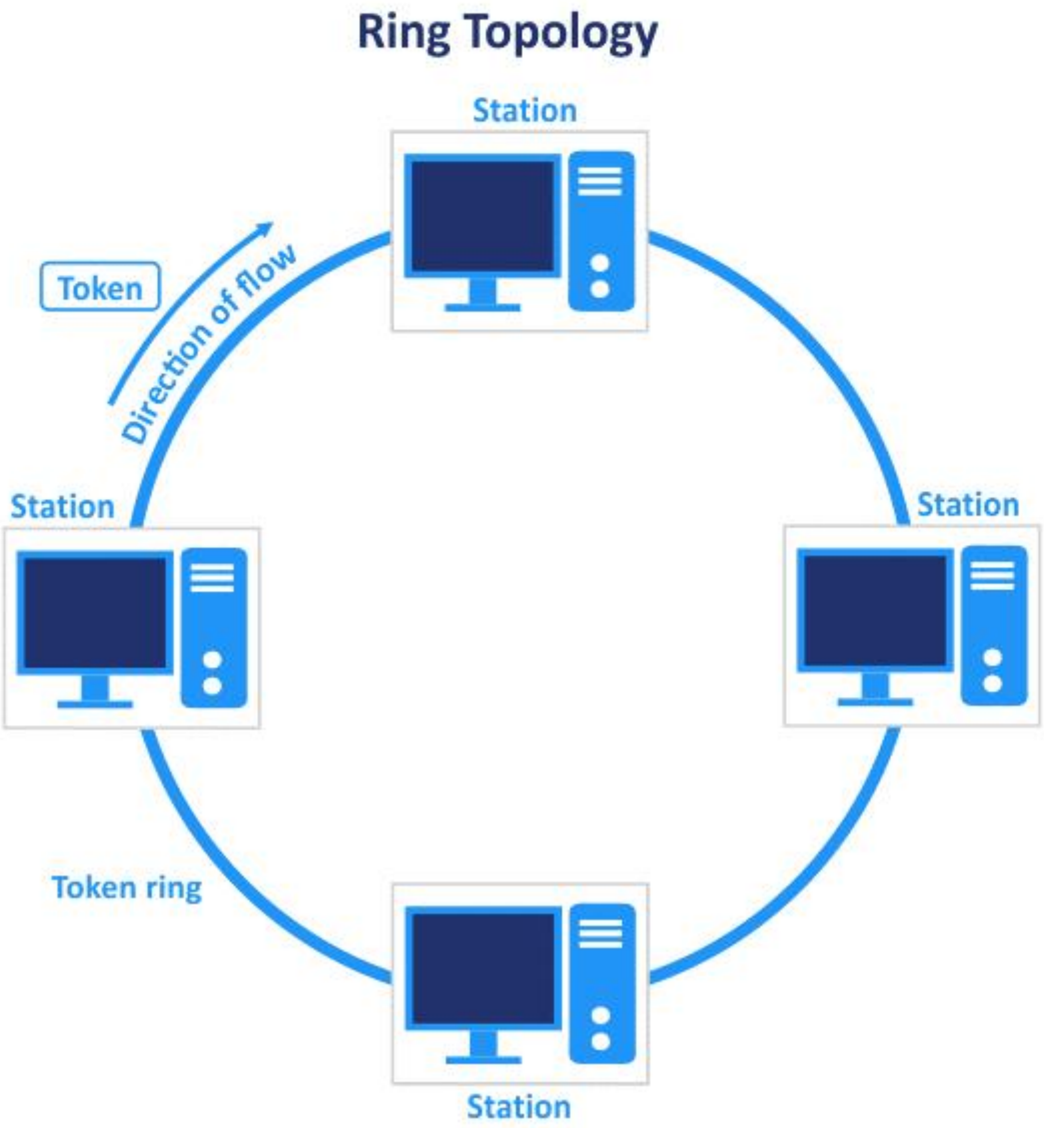
- Devices are connected in a circular path.

Advantages

•Organized communication

Disadvantages

•Failure of one node can disrupt network



Basic IoT Network Topologies

D. Mesh Topology

- Each node connects to multiple nodes.

Advantages

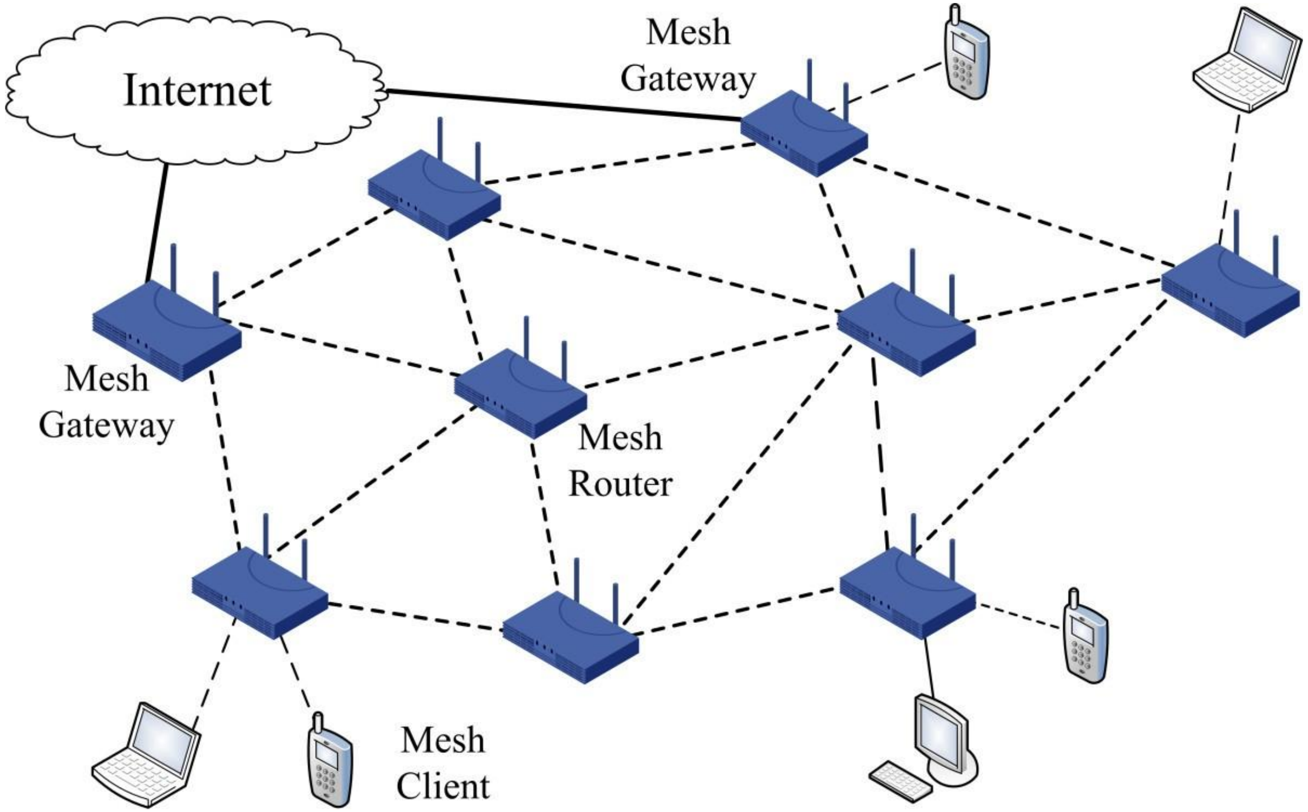
- High reliability
- Self-healing network

Disadvantages

- Complex configuration
- Higher cost

Example

- ZigBee-based smart home systems.



Basic IoT Network Topologies

E. Tree Topology

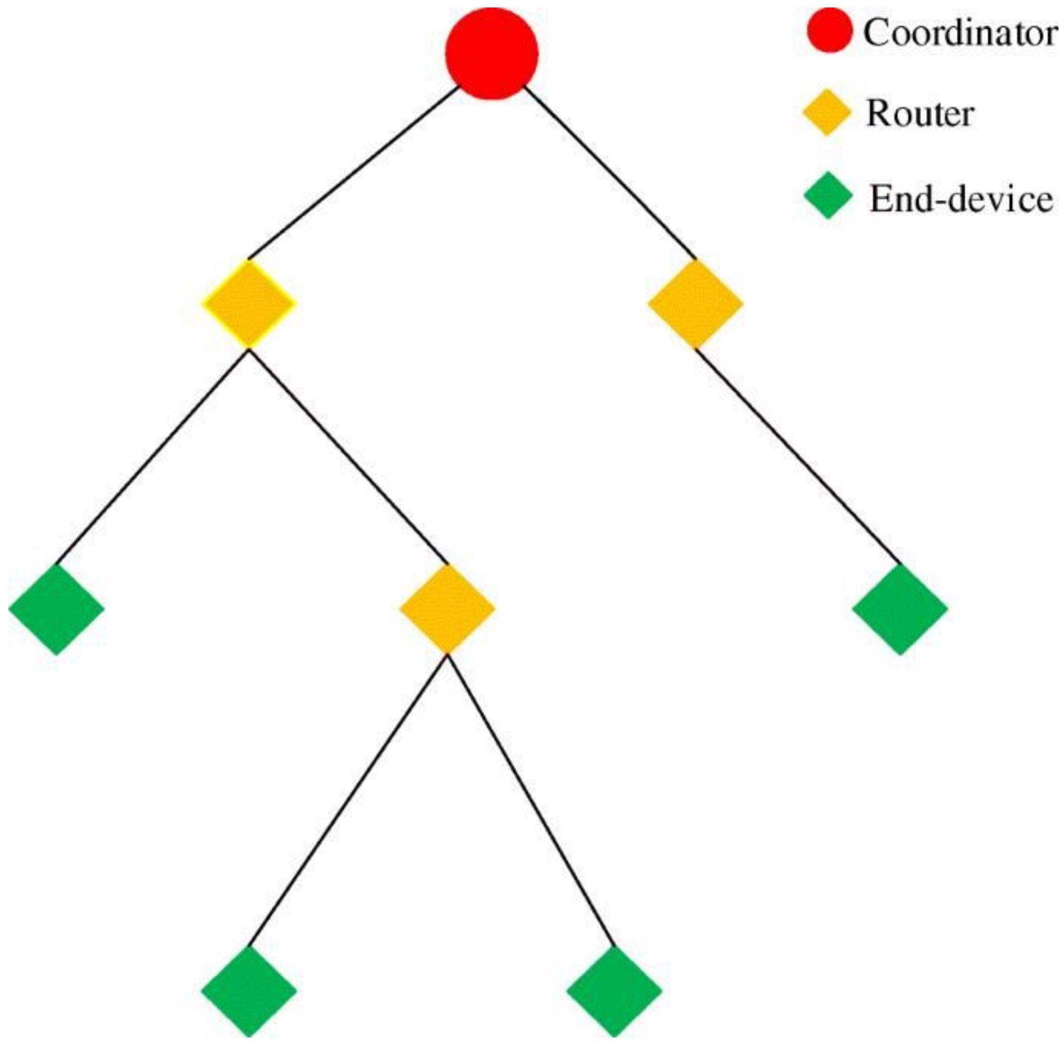
- Combination of star and bus topologies in
- hierarchical form.

Advantages

- Easy expansion
- Scalable

Disadvantages

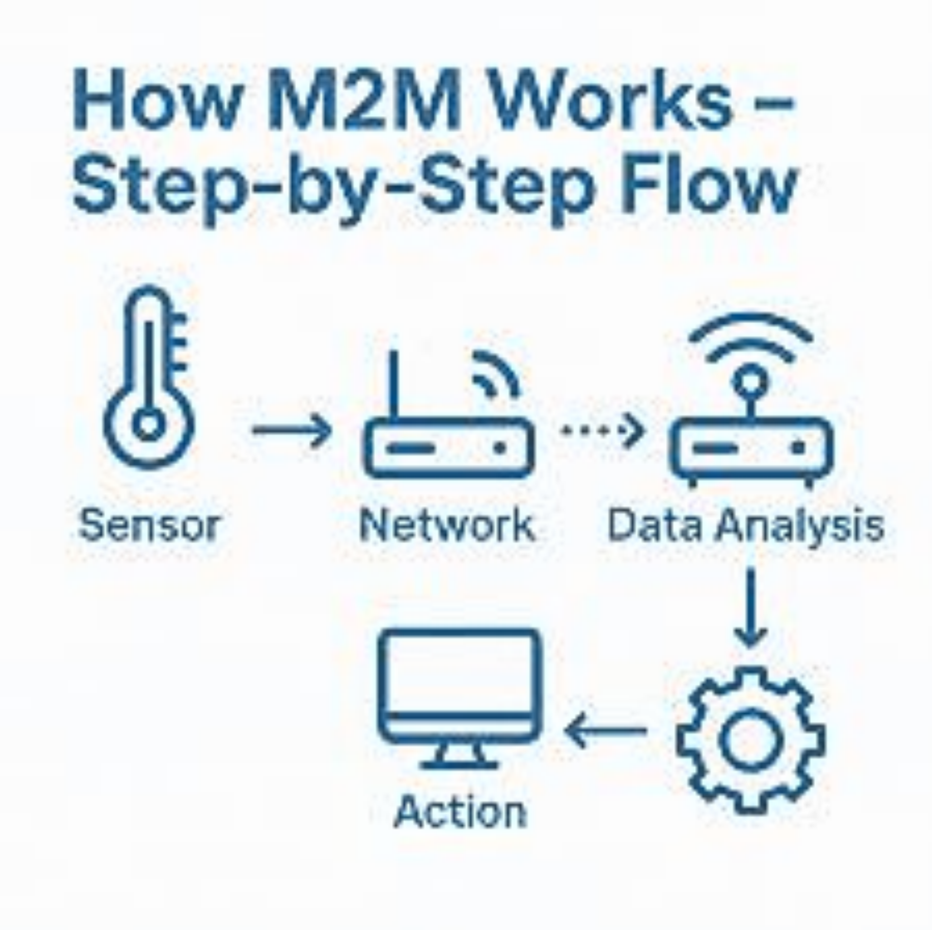
- Dependency on backbone network



Introduction to M2M (Machine-to-Machine)

Machine-to-Machine (M2M) communication refers to direct communication between machines or devices without human intervention.

Features
- Automatic data exchange
- Uses wired or wireless networks
- Limited Internet involvement
- Device-to-device communication



Examples
- ATM Machines
- Vending Machines
- Industrial Automation
- Smart Meters

M2M Value Chain

- The M2M Value Chain describes the flow of data from devices to business applications.

Components of M2M Value Chain

1. Devices
Sensors
RFID Tags
Smart Meters

2. Communication Network
GSM
Wi-Fi
Bluetooth
ZigBee

3. M2M Gateway
Collects and processes data
Connects devices to servers

4. Data Processing
Platform
Stores and analyzes data

5. Business Applications
Monitoring
Reporting
Decision Making

Flow

Device → Network → Gateway → Server → Application

Difference Between IoT and M2M

Feature	IoT	M2M
Full Form	Internet of Things	Machine-to-Machine
Connectivity	Internet Based	Direct Communication
Communication	Many-to-Many	Point-to-Point
Network	IP Networks	Cellular/Wired Networks
Scalability	Highly Scalable	Limited
Data Storage	Cloud-Based	Local Servers
User Interaction	High	Low
Protocols	MQTT, CoAP, HTTP	GSM, Modbus
Applications	Smart Home, Smart City	ATM, Vending Machine

Software Defined Network (SDN)

- Software Defined Networking (SDN) is a networking approach where the control of the network is separated from the hardware devices.

Main Components

1. Application Layer

- Network applications
- Security applications

2. Control Layer

- SDN Controller
- Makes routing decisions

3. Infrastructure Layer

- Switches
- Routers
- IoT Devices

Traditional Network

Control Plane + Data Plane inside each switch/router

SDN Network

SDN Controller

Switch Switch Switch

SDN for IoT

- IoT networks consist of millions of connected devices. Managing them becomes difficult using traditional networks.
- SDN for IoT provides centralized control and management of IoT devices.

Architecture

IoT Devices

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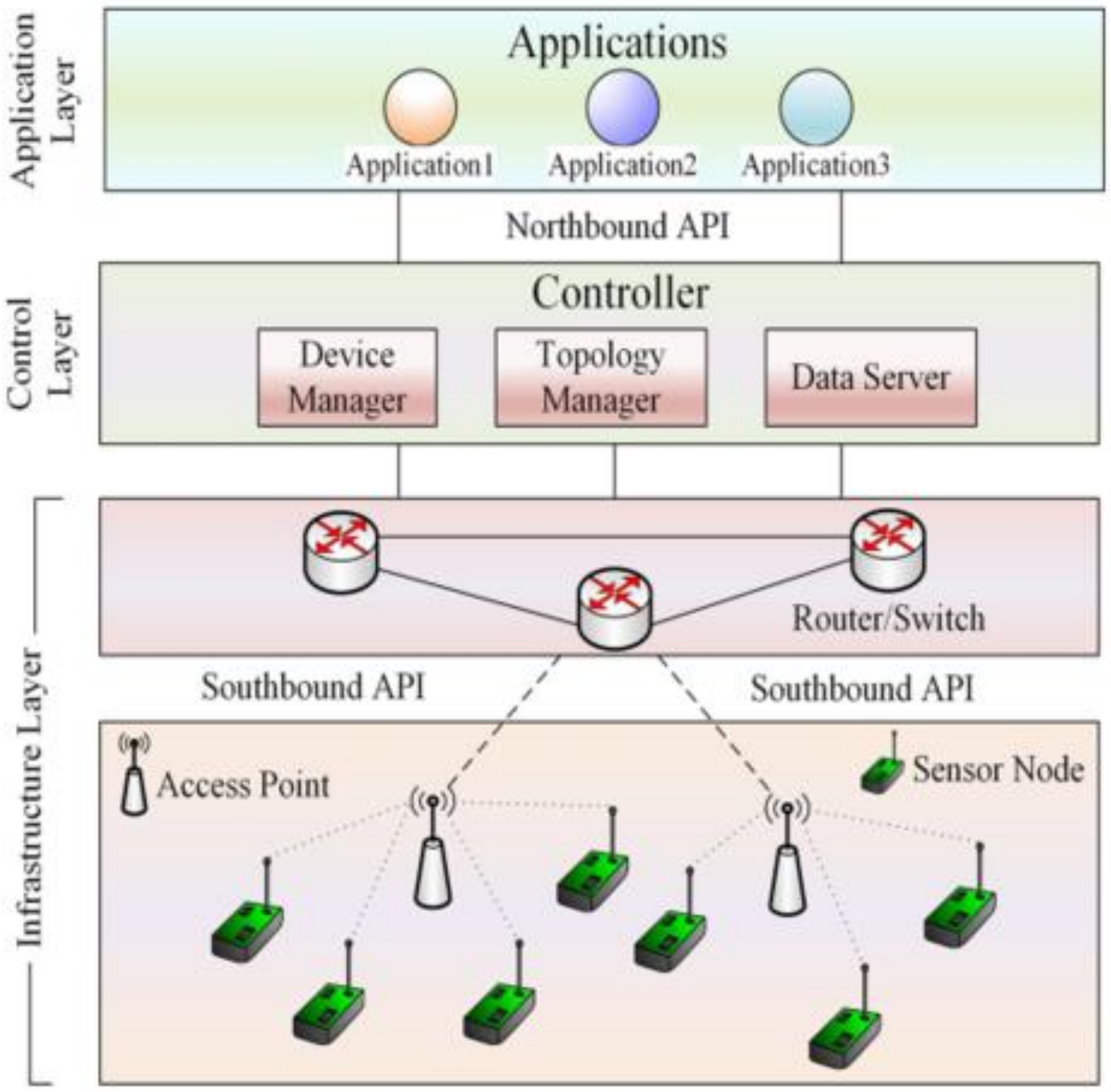
Network Switches

↓

SDN Controller

↓

Cloud Applications



Benefits of SDN in IoT

1. Centralized Control

- Single controller manages all devices.

2. Better Security

- Detects malicious traffic.
- Controls network access.

3. Scalability

- Supports millions of IoT devices.

4. Traffic Management

- Optimizes data flow.

5. Flexibility

- Network behavior can be changed through software.

Applications

- Smart Cities
- Smart Healthcare
- Smart Transportation
- Industrial IoT
- Smart Grid Systems

Security Challenges in IoT

- Security refers to protecting IoT devices, networks, and data from unauthorized access and cyberattacks.

1. Unauthorized Access

- Hackers may gain control of IoT devices.
- Weak passwords increase vulnerability.

2. Data Breaches

- Sensitive information may be stolen during transmission.

3. Malware Attacks

- Viruses and malicious software can infect devices.

4. Denial of Service (DoS) Attacks

- Attackers overload devices or networks, making services unavailable.

5. Physical Attacks

- Attackers may physically access IoT devices and manipulate them.

Security Solutions

- Strong authentication
- Data encryption
- Regular software updates
- Secure communication protocols
- Firewalls and intrusion detection systems

Privacy Challenges in IoT

- Privacy concerns the protection of personal and sensitive information collected by IoT devices.

Data Collection

- IoT devices continuously collect user information.

Data Sharing

- Information may be shared without user knowledge.

Location Tracking

- Smart devices can track user movements.

User Profiling

- Personal behavior patterns can be analyzed.

Privacy Protection Measures

- User consent mechanisms
- Data minimization
- Anonymization techniques
- Secure storage systems

Trust Challenges in IoT

- Trust is the confidence that users have in IoT devices and systems.

Device Reliability

- Devices may provide incorrect data.

Data Integrity

- Data can be modified during transmission.

System Transparency

- Users may not know how data is used.

Third-Party Services

- Cloud providers may not always be fully trusted.

Building Trust

- Secure authentication
- Digital certificates
- Transparent policies
- Reliable device management

Interoperability Challenges in IoT

- Interoperability is the ability of different devices and systems to communicate and work together.

Different Communication Protocols

- Wi-Fi
- Bluetooth
- ZigBee
- LoRaWAN

Different Manufacturers

- Devices use different standards.

Data Format Differences

- Systems may store data in different formats.

Platform Compatibility

- Applications may not support all devices.

Solutions

- Common standards
- Open APIs
- Middleware platforms
- Standard communication protocols

Device-Related Issues in IoT

- IoT devices often have limited resources and capabilities.

Limited Processing Power
- Small processors cannot perform complex tasks.

Limited Memory
- Storage capacity is often restricted.

Power Constraints
- Battery-operated devices require low power consumption.

Hardware Failures
- Sensors may become damaged or inaccurate.

Firmware Updates
- Updating millions of devices is difficult.

Scalability Issues
- Managing large numbers of devices is challenging.

Solutions
- Energy-efficient hardware
- Remote device management
- Edge computing
- Automated firmware updates

IoT Standardization

- IoT Standardization refers to developing common rules and protocols that ensure compatibility and interoperability among IoT devices and platforms.

Need for Standardization

- Seamless communication
- Improved security
- Better interoperability
- Vendor independence
- Reduced development costs

Areas of Standardization

Communication Standards

- Wi-Fi
- Bluetooth
- ZigBee
- LoRaWAN

Data Standards

- Standard data formats
- Common APIs

Security Standards

- Authentication methods
- Encryption techniques

Device Management Standards

- Remote monitoring
- Firmware updates

Organizations Involved in IoT Standardization

IEEE

- Develops networking and communication standards.

IETF

- Develops Internet protocols such as IPv6 and CoAP.

ITU

- Creates global telecommunication standards.

ISO

- Develops international technology standards.

oneM2M

- Provides standards for M2M and IoT systems.